A Head Camera Is (Almost) All You Need

An electronics-free UMI glove labelled and observed by one moving camera

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Why a head camera only

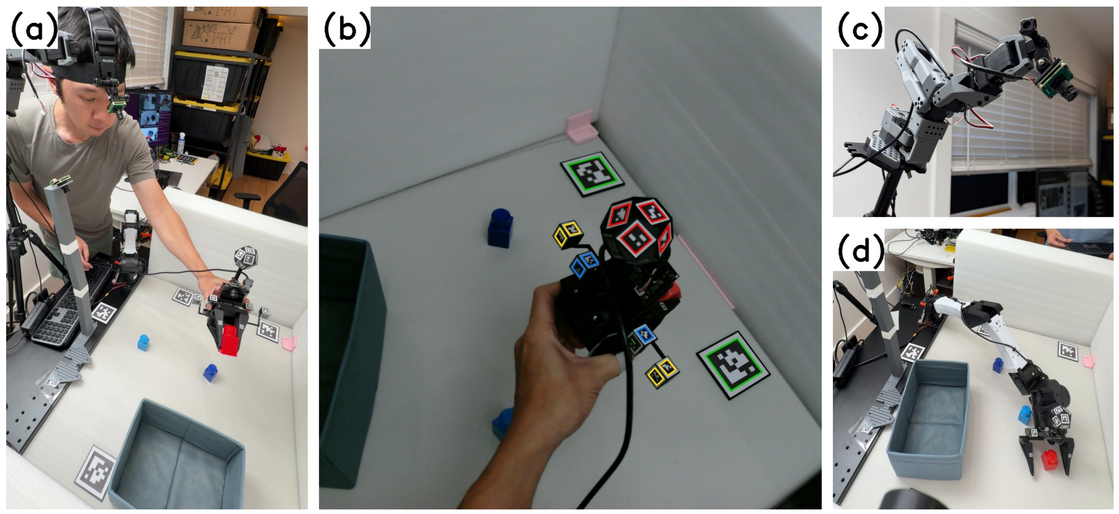
- Wrist devices need a **second stream** to track the gripper: SLAM, an IMU, a controller or an external camera.
- **Gaze leads the hand**: the gripper stays in view, at arm's length, with the goal.
- The frames the policy sees are the frames that **carry the labels**.

No SLAM, no IMU, no wrist camera, no cables.

$144 glove, camera, strap

103 g on the head

One head camera: the policy's observation and the pose labels



(a) A $122 4K head camera, a $22 electronics-free glove, four paper tiles. (b) The camera's own frame: tiles give the camera pose, the ball the gripper pose, the jaw tags the opening. (c) At deployment the same camera rides a $122 six-joint arm (d) beside the robot.

Label accuracy

- **Camera pose** from the paper tiles, refit from the footage to 0.5 px.
- **Gripper pose** by PnP on a 15 mm ArUco dodecahedron.
- **Jaw width** from printed tails on the jaws.

6.5 mm tool point

1.6° orientation

3.5 % jaw width, of stroke

Median error against the robot's encoders; a label on 93–100 % of frames.

Collection

2–3× demos per minute vs teleop

198 demos in four sessions

6–8 s per demo; teleop 14 s

0 robot time

Continuous recording, episodes cut by a keypress; 0.7–7.4 % of frames dropped for tracking.

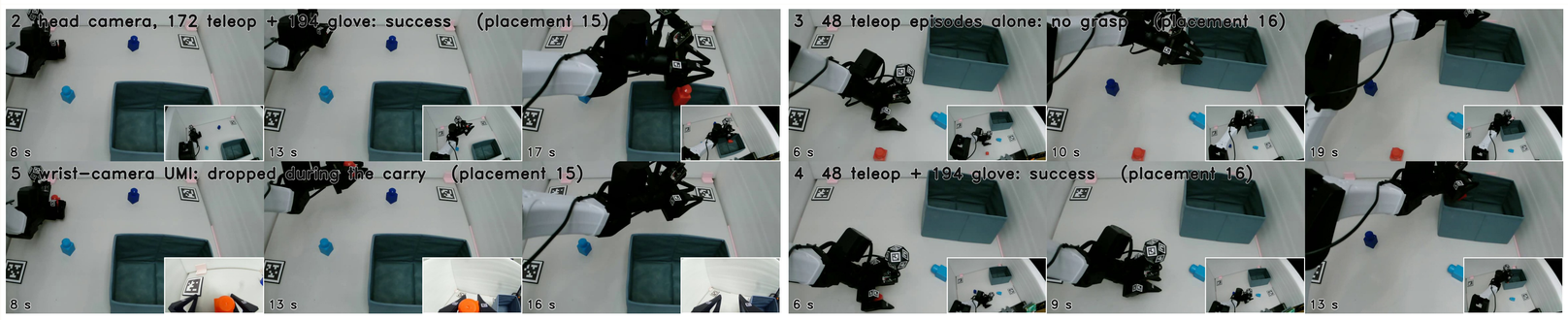
Deployment

- The same camera on a $122 six-joint arm beside the YAM.
- It follows the gripper **from the robot's encoders alone**, in teleoperation and in rollouts, so the policy sees what the collector saw.

46 mm head pose vs the collector

5.4° static camera: 87 mm, 7.0°

Four rollouts



Fixed workspace camera, each policy's own view inset. Left, placement 15: the head-camera policy carries the block to the bin; the wrist-camera policy, seeing only block and wall, drops it mid-carry. Right, placement 16: 48 teleoperated episodes alone never grasp; 48 plus 194 glove demos succeed.

Glove demos improve a full teleop set

Eight ACT policies, joint-space actions. Rollouts: 25 placements, one trial each, 60 s. Every co-trained mix uses the same 194 glove demos.

18/25 172 teleop episodes

20/25 172 teleop + 194 glove

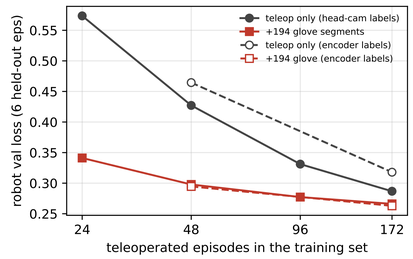
Validation loss on held-out teleop episodes 0.318 → 0.263.

. . . and stretch a small one

10/25 48 teleop

17/25 48 teleop + 194 glove

18/25 172 teleop, no glove



Loss 0.464 → 0.295. On the teleop-only curve the glove set is worth **65–100 teleop episodes**: 2–3 glove demos per teleoperated one. The gain shrinks as the teleop budget grows.

Head-camera labels replace the encoders

0.318 teleop labelled by encoders

0.328 teleop labelled by the head camera

Same 172 episodes, same held-out targets: 3 % of loss. On the robot 18/25 vs 15/25, within noise ($p = 0.58$).

The view decides whether the task completes

20/25 head camera

15/25 wrist fisheye (UMI)

14/25 wrist fisheye, head labels

12/25 fixed camera

All four co-trained, offline loss within 0.011 of each other. Head-camera policies fail by never grasping; wrist-camera policies **drop the block mid-carry** (12 of 21 failures vs 1 of 45), every time while the fisheye looked away from the bin.

It is the view, not the labels

- Same wrist view, labels from the fixed camera or from the head camera: **15/25 vs 14/25**, no change.
- Same head labels, wrist view or head view: **14/25 → 20/25**, and the drops disappear.

Limits and next

One task, one collector, one tile layout. Glove-only policies do not transfer: co-training is required. Next: bimanual, tile-free in the wild.